

Contents

Education	2
Research Experience	2
Publications	2
Refereed Conference & Journal Articles	2
Preprints	2
Presentations	3
Software	3

Education

2023-Present	PhD in Computational Science, Engineering, and Mathematics Oden Institute, University of Texas at Austin; Austin, TX Advisor: George Biros
2023-2026	MS in Computational Science, Engineering, and Mathematics Oden Institute, University of Texas at Austin; Austin, TX
2019-2023	BSLAS in Mathematics & Computer Science University of Illinois at Urbana-Champaign; Champaign, IL

Research Experience

Publications

Refereed Conference & Journal Articles

- [A1] **G. Kosmacher**, D. Max, Z. Rapti, C. E. Cáceres, and T. E. Stewart Merrill, “The role of host immunity and the environment in seasonal disease dynamics,” *Bulletin of Mathematical Biology*, vol. 88, no. 2, p. 18, Jan. 2026. DOI: [10.1007/s11538-025-01582-3](#).
- [A2] **G. Kosmacher** and K. Shankari, “Evaluating the interplay between trajectory segmentation and mode inference error,” *Transportation Research Record: Journal of the Transportation Research Board*, vol. 2678, no. 7, pp. 32–49, Jul. 2024. DOI: [10.1177/03611981231208154](#).
- [A3] Y. Du, **G. Kosmacher**, Y. Liu, J. Massman, J. Palmer, T. Thieme, J. Wu, and Z. Zhang, “Packing densities of delzant and semitoric polygons,” *SIGMA. Symmetry, Integrability and Geometry: Methods and Applications*, vol. 19, p. 081, Oct. 2023. DOI: [10.3842/SIGMA.2023.081](#).

Preprints

- [PP1] **G. Kosmacher**, Z. Du, J. Bagge, and G. Biros, “A performance portable fast ewald summation for stokes flow,” Jun. 2026. arXiv: [2606.19059 \[math.NA\]](#).
- [PP2] **G. Kosmacher**, E. T. Phipps, and S. Rajamanickam, “A performance portable matrix free dense mttkrp in genten,” preprint, Oct. 2025. arXiv: [2510.14891 \[cs.MS\]](#).

Presentations

Software